

Selected Topics Of Living Systems HW2

Daniel Haim Breger, 316136944

Technion

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1 Linear Stability and Waves of Dry Active Polar Matter

The dynamics of dry polar active matter is given by

$$\frac{\partial \rho}{\partial t} = -\nabla \cdot \left(-\gamma_\rho \nabla \frac{\delta F}{\delta \rho} + v_0 \rho \mathbf{p} \right) \quad (1)$$

$$\frac{\partial \mathbf{p}}{\partial t} + \lambda (\mathbf{p} \cdot \nabla) \mathbf{p} = -\gamma_p \frac{\delta F}{\delta \mathbf{p}} \quad (2)$$

We focus on the polar phase in 2D with $\rho = \rho_0 + \delta\rho(\mathbf{r})$ and $\mathbf{p} = \hat{\mathbf{x}} + \delta p(\mathbf{r})\hat{\mathbf{y}}$ where $|\delta\rho| \ll \rho_0$ and $|\delta p| \ll 1$ and consider the free energy

$$F = \int d^2r \left[\frac{\chi}{2} \left(\frac{\delta\rho}{\rho_0} \right)^2 - \frac{1}{2} h_\parallel p_\alpha p_\alpha + \frac{K}{2} \partial_\alpha p_\beta \partial_\alpha p_\beta - \omega_1 \frac{\delta\rho}{\rho_0} \nabla \cdot \mathbf{p} \right] \quad (3)$$

Here, we have used the summation convention. Note that $h_\parallel$ is a Lagrange multiplier to ensure that $p_\alpha p_\alpha = 1$. In class, we considered $\delta p(\mathbf{r}) = \delta p(y)$ and $\omega_1 > 0$. Perform a linear stability analysis and find the dispersion relation for (a) $\delta p = \delta p(y)$ and $\omega_1 < 0$ (b) general $\delta p(\mathbf{r})$ and $\omega_1 > 0$. In which case is the system unstable? Explain why. When is it stable - what is the velocity of propagating waves?

Before starting the specific cases, I'll rewrite the free energy using the known relations for the density and polarization.

$$p_\alpha p_\alpha = \mathbf{p} \cdot \mathbf{p} = 1 + (\delta p(\mathbf{r}))^2 \quad (4)$$

$$\partial_\alpha p_\beta \partial_\alpha p_\beta = (\partial_x \delta p(\mathbf{r}))^2 + (\partial_y \delta p(\mathbf{r}))^2 \quad (5)$$

$$\nabla \cdot \mathbf{p} = \partial_y \delta p(\mathbf{r}) \quad (6)$$

$$\frac{\delta\rho}{\rho_0} = \frac{\delta\rho(\mathbf{r})}{\rho_0} \quad (7)$$

now I can insert this into the free energy

$$F = \int d^2r \left[\frac{\chi}{2} \left(\frac{\delta\rho}{\rho_0} \right)^2 - \frac{1}{2} h_\parallel p_\alpha p_\alpha + \frac{K}{2} \partial_\alpha p_\beta \partial_\alpha p_\beta - \omega_1 \frac{\delta\rho}{\rho_0} \nabla \cdot \mathbf{p} \right] \quad (8)$$

$$= \int d^2r \left[\frac{\chi}{2} \left(\frac{\delta\rho(\mathbf{r})}{\rho_0} \right)^2 - \frac{1}{2} h_\parallel \left(1 + (\delta p(\mathbf{r}))^2 \right) + \frac{K}{2} (\partial_x \delta p(\mathbf{r}))^2 + (\partial_y \delta p(\mathbf{r}))^2 - \omega_1 \frac{\delta\rho(\mathbf{r})}{\rho_0} \partial_y \delta p(\mathbf{r}) \right] \quad (9)$$

next we find

$$\frac{\delta F}{\delta \rho} = \chi \frac{\delta\rho(\mathbf{r})}{\rho_0^2} - \frac{\omega_1}{\rho_0} \partial_y \delta p(\mathbf{r}) \quad (10)$$

to do the other variation I'll look at it term by term for $p_\alpha \rightarrow p_\alpha + \varepsilon \eta_\alpha$

$$\delta(p_\alpha p_\alpha) = p_\alpha^2 + 2\varepsilon p_\alpha \eta_\alpha + \varepsilon^2 \eta_\alpha^2 = 2p_\alpha \eta_\alpha \quad (11)$$

$$\delta(\partial_\alpha p_\beta \partial_\alpha p_\beta) = 2\partial_\alpha p_\beta \partial_\alpha (\delta p_\beta) \quad (12)$$

which needs an integration by parts

$$\int d^2r 2\partial_\alpha p_\beta \partial_\alpha (\delta p_\beta) = - \int d^2r 2\partial_\alpha \partial_\alpha p_\beta \delta p_\beta = - \int d^2r 2\nabla^2 p_\beta \delta p_\beta \quad (13)$$

with the boundary terms cancelling out because it's a variation. And finally

$$\delta \left(\frac{\delta \rho}{\rho_0} \nabla \cdot \mathbf{p} \right) = \frac{\delta \rho}{\rho_0} \partial_\alpha \delta p_\alpha \quad (14)$$

integrating by parts

$$\int d^2r \frac{\delta \rho}{\rho_0} \partial_\alpha \delta p_\alpha = - \int d^2r \partial_\alpha \left(\frac{\delta \rho}{\rho_0} \right) \delta p_\alpha \quad (15)$$

connecting everything back

$$\frac{\delta F}{\delta \mathbf{p}} = -h_\parallel \mathbf{p} - K \nabla^2 \mathbf{p} + \omega_1 \nabla \left(\frac{\delta \rho}{\rho_0} \right) \quad \frac{\delta F}{\delta \rho} = \chi \frac{\delta \rho(\mathbf{r})}{\rho_0^2} - \frac{\omega_1}{\rho_0} \partial_y \delta p(\mathbf{r}) \quad (16)$$

Looking at the dynamic equation for density

$$\frac{\partial \rho}{\partial t} = -\nabla \cdot \left(-\gamma_\rho \nabla \frac{\delta F}{\delta \rho} + v_0 \rho \mathbf{p} \right) \quad (17)$$

$$= \gamma_\rho \nabla^2 \left(\frac{\delta F}{\delta \rho} \right) - \nabla \cdot (v_0 \rho \mathbf{p}) \quad (18)$$

inserting the density and assuming ρ_0 is constant in time

$$\frac{\partial \delta \rho(\mathbf{r})}{\partial t} = \gamma_\rho \nabla^2 \left(\frac{\delta F}{\delta \rho} \right) - \nabla \cdot (v_0 \rho \mathbf{p}) \quad (19)$$

where

$$\nabla \cdot (v_0 (\rho_0 + \delta \rho(\mathbf{r})) \mathbf{p}) \simeq v_0 (\partial_x \delta \rho(\mathbf{r}) + \rho_0 \partial_y \delta p(\mathbf{r})) \quad (20)$$

where we omitted the term quadratic in delta. Therefore

$$\partial_t \delta \rho(\mathbf{r}) = \gamma_\rho \nabla^2 \left(\chi \frac{\delta \rho(\mathbf{r})}{\rho_0^2} - \frac{\omega_1}{\rho_0} \partial_y \delta p(\mathbf{r}) \right) - v_0 (\partial_x \delta \rho(\mathbf{r}) + \rho_0 \partial_y \delta p(\mathbf{r})) \quad (21)$$

At this point I realized I could have linearized things about 10 equations ago. A previous version for this question was 7 pages long... Moving on, to first order this is

$$\partial_t \delta \rho(\mathbf{r}) = -v_0 (\partial_x \delta \rho(\mathbf{r}) + \rho_0 \partial_y \delta p(\mathbf{r})) \quad (22)$$

rearranging

$$\partial_t \delta \rho + v_0 \partial_x \delta \rho + v_0 \rho_0 \partial_y \delta p(\mathbf{r}) = 0 \quad (23)$$

The equation for polarity is

$$\frac{\partial \mathbf{p}}{\partial t} + \lambda (\mathbf{p} \cdot \nabla) \mathbf{p} = -\gamma_p \frac{\delta F}{\delta \mathbf{p}} \quad (24)$$

inserting the delta we have

$$\frac{\partial \mathbf{p}}{\partial t} + \lambda (\mathbf{p} \cdot \nabla) \mathbf{p} = -\gamma_p \left(-h_{\parallel} \mathbf{p} - K \nabla^2 \mathbf{p} + \omega_1 \nabla \left(\frac{\delta \rho}{\rho_0} \right) \right) \quad (25)$$

dropping the non-linear terms

$$\frac{\partial \mathbf{p}}{\partial t} + \lambda (\mathbf{p} \cdot \nabla) \mathbf{p} = -\gamma_p \omega_1 \nabla \left(\frac{\delta \rho}{\rho_0} \right) \quad (26)$$

rewriting with indices

$$\partial_t \delta p_{\alpha} + \lambda p_{\beta} \partial_{\beta} \delta p_{\alpha} = -\gamma_p \frac{\omega_1}{\rho_0} \partial_{\alpha} \delta \rho \quad (27)$$

but since $\partial_x \delta \rho = 0$ and $\delta p_x = 0$

$$\partial_t \delta p(y) = -\gamma_p \frac{\omega_1}{\rho_0} \partial_y \delta \rho \quad (28)$$

taking the derivative of the density equation

$$\partial_t \partial_t \delta \rho + v_0 \partial_t \partial_x \delta \rho + v_0 \rho_0 \partial_t \partial_y \delta p(\mathbf{r}) = 0 \quad (29)$$

we can replace the order of differentiation so we insert the expression we got from the polarization equation

$$\partial_t \partial_t \delta \rho + v_0 \partial_t \partial_x \delta \rho - v_0 \rho_0 \partial_y \gamma_p \frac{\omega_1}{\rho_0} \partial_y \delta \rho = 0 \quad (30)$$

rearranging

$$(\partial_{tt} + v_0 \partial_{tx}) \delta \rho = \gamma_p v_0 \omega_1 \partial_{yy} \delta \rho \quad (31)$$

but $\partial_x \delta \rho = 0$

$$\partial_{tt} \delta \rho = \gamma_p v_0 \omega_1 \partial_{yy} \delta \rho \quad (32)$$

which is a wave equation, thus with a standard wave ansatz we find

$$\omega^2 = \gamma_p v_0 \omega_1 k^2 \quad (33)$$

Therefore

$$\omega = \pm k \sqrt{\gamma_p v_0 \omega_1} \quad (34)$$

since $\omega_1 < 0$, ω is imaginary. If $\Im(\omega) < 0$, any created waves will decay and the system as a whole will be stable. Otherwise, waves will grow exponentially, meaning the system is unstable. Since the dispersion relation is linear in k , the phase velocity and group velocity are the same and are

$$c = \sqrt{\gamma_p v_0 \omega_1} \quad (35)$$

2 Nematic Order

1. The nematic tensor in the fully ordered state is given by $Q_{\alpha\beta} = n_\alpha n_\beta - \delta_{\alpha\beta}/d$ where n_α is the director field and d is the dimension. Show that in 2D

$$\mathbf{Q} = \frac{1}{2} \begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix} \quad (36)$$

where $\mathbf{n} = (\cos \theta, \sin \theta)$.

2. Consider the Frank free energy

$$F = \int d\mathbf{r} \left[\frac{1}{2} K_1 (\nabla \cdot \mathbf{n})^2 + \frac{1}{2} K_2 [\mathbf{n} \cdot (\nabla \times \mathbf{n})]^2 + \frac{1}{2} K_3 [\mathbf{n} \times (\nabla \times \mathbf{n})]^2 \right] \quad (37)$$

Show that within the one-constant approximation ($K_i = K$), the free energy reduces to

$$F = \int d\mathbf{r} \frac{1}{2} K \partial_\alpha n_\beta \partial_\alpha n_\beta + F_b \quad (38)$$

where F_b is a boundary term, called saddle-splay. In 2D, show that this free energy can also be written as $F = \int d\mathbf{r} \frac{1}{2} K (\nabla \theta)^2$.

2.1 Nematic Tensor

By a direction calculation, in 2D

$$Q_{\alpha\beta} = n_\alpha n_\beta - \delta_{\alpha\beta}/2 \quad (39)$$

$$= \begin{pmatrix} \cos^2 \theta - \frac{1}{2} & \cos \theta \sin \theta \\ \sin \theta \cos \theta & \sin^2 \theta - \frac{1}{2} \end{pmatrix} \quad (40)$$

$$= \frac{1}{2} \begin{pmatrix} 2 \cos^2 \theta - 1 & 2 \cos \theta \sin \theta \\ 2 \sin \theta \cos \theta & 2 \sin^2 \theta - 1 \end{pmatrix} \quad (41)$$

$$= \frac{1}{2} \begin{pmatrix} 2 \cos^2 \theta - \sin^2 \theta - \cos^2 \theta & 2 \cos \theta \sin \theta \\ 2 \sin \theta \cos \theta & 2 \sin^2 \theta - \sin^2 \theta - \cos^2 \theta \end{pmatrix} \quad (42)$$

$$= \frac{1}{2} \begin{pmatrix} \cos^2 \theta - \sin^2 \theta & 2 \cos \theta \sin \theta \\ 2 \sin \theta \cos \theta & \sin^2 \theta - \cos^2 \theta \end{pmatrix} \quad (43)$$

$$= \frac{1}{2} \begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix} \quad (44)$$

where the last transition used the trig identities

$$\sin 2\theta = 2 \sin \theta \cos \theta \quad \cos 2\theta = \cos^2 \theta - \sin^2 \theta \quad (45)$$

2.2 Frank Free Energy

Since we are working within the one-constant approximation, we can immediately simplify the free energy to

$$F = \int d\mathbf{r} \frac{K}{2} \left[(\nabla \cdot \mathbf{n})^2 + [\mathbf{n} \cdot (\nabla \times \mathbf{n})]^2 + [\mathbf{n} \times (\nabla \times \mathbf{n})]^2 \right] \quad (46)$$

if we define for simplicity $\mathbf{a} = \nabla \times \mathbf{n}$ then

$$F = \int d\mathbf{r} \frac{K}{2} \left[(\nabla \cdot \mathbf{n})^2 + [\mathbf{n} \cdot \mathbf{a}]^2 + [\mathbf{n} \times \mathbf{a}]^2 \right] \quad (47)$$

but

$$(\mathbf{n} \times \mathbf{a})^2 = (\varepsilon_{ijk} n_j a_k) (\varepsilon_{lmn} n_m a_n) = \varepsilon_{ijk} \varepsilon_{lmn} n_j a_k n_m a_n \quad (48)$$

$$= (\delta_{jm} \delta_{kn} - \delta_{jn} \delta_{mk}) n_j n_m a_k a_n \quad (49)$$

$$= n_j n_j a_k a_k - n_j n_k a_j a_k \quad (50)$$

$$= n^2 a^2 - (\mathbf{n} \cdot \mathbf{a})^2 \quad (51)$$

therefore

$$F = \int d\mathbf{r} \frac{K}{2} \left[(\nabla \cdot \mathbf{n})^2 + n^2 a^2 \right] \quad (52)$$

but since $\mathbf{n}$ is a unit vector

$$F = \int d\mathbf{r} \frac{K}{2} \left[(\nabla \cdot \mathbf{n})^2 + (\nabla \times \mathbf{n})^2 \right] \quad (53)$$

converting to index notation

$$F = \int d\mathbf{r} \frac{K}{2} [\partial_i n_i \partial_j n_j + \partial_j n_k \partial_j n_k - \partial_j n_k \partial_k n_j] \quad (54)$$

$$= \int d\mathbf{r} \frac{K}{2} [\partial_i n_j \partial_i n_j - \partial_i (n_j \partial_j n_i - n_i \partial_j n_j)] \quad (55)$$

if we identify $F_b = -\partial_i (n_j \partial_j n_i - n_i \partial_j n_j)$ we found what's needed. If we now assume we are in 2D then

$$\mathbf{n} = (\cos \theta, \sin \theta) \quad (56)$$

therefore

$$\partial_\alpha \mathbf{n} = \partial_\theta \mathbf{n} \partial_\alpha \theta = (-\sin \theta, \cos \theta) \partial_\alpha \theta \quad (57)$$

and thus

$$(\partial_\alpha \mathbf{n})^2 = (\sin^2 \theta + \cos^2 \theta) (\partial_\alpha \theta)^2 = (\partial_\alpha \theta)^2 = (\nabla \theta)^2 \quad (58)$$

which gives us

$$\int d\mathbf{r} \frac{K}{2} (\nabla \theta)^2 \quad (59)$$